Object Detection for Retail Product Recognition

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2024 9th International Conference on Business and Industrial Research (ICBIR) | 979-8-3503-8302-7/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICBIR61386.2024.10875711

***Abstract*—The retail sector is a vital driver of economic growth. The retail industry must adopt technology to enhance productivity, streamline operations, and minimize human errors in order to continue its crucial economic role. During the COVID-19 pandemic, the purchasing power volumes globally grew from February 2020 to April 2021, and the retail sector gained 35 percents in market capitalization. This data highlights a compelling research opportunity. Consequently, artificial intelligence (AI) has become a focus of significant interest and has widely adopted technology, which includes computer vision to recognize and detect retail products. This paper aims to explore YOLO's performance in retail product recognition. The research compares different YOLO versions to identify on-shelf grocery items. In this research, YOLOv8 is applied and divided into five subcategories: YOLOv8n (nano), YOLOv8s (small), YOLOv8m (medium), YOLOv8l (large), and YOLOv8x (extra-large). They were evaluated with Grozi-120 and SKU110K datasets. The evaluation metrics are precision, recall, mAP50, and mAP50-95. The result shows that YOLOv8x provides the best overall performance in both datasets, where mAP50 metrics exhibits the highest score at 92.6 percents in SKU110K. The outcomes show that the YOLOv8 model works well for retail product detection.**

Keywords—object detection, computer vision, machine learning, deep learning, convolutional neutral network

I. Introduction

The retail industry is a crucial driver of economic development. Technology adoption is essential for the retail sector to enhance traditional processes, improve productivity, and minimize individual errors, ensuring continued economic vitality. The advancement in technologies is forcing substantial innovation in the retail sector. The change has rapidly risen during the Covid-19 crisis. During the pandemic, the spending volumes globally grew from February 2020 to April 2021, and the retail sector gained 35% in market capitalization [1]. Furthermore, adopting technology in the retail business escalates customer attention and expectations. A recent study [2] revealed that convenience, level of modernization, and in-store digitization technologies are the considerable factors that affect customers' store preferences and selections. Moreover, the expectation of global spending on artificial intelligence (AI) would dramatically increase up to $12 billion in 2023 from $3.6 billion in 2019, or over 300% increase [3].

Therefore, to maintain competitiveness, identifying a product on a retail store shelf is an essential activity, which is a common individual skill, but it is prone to errors. Relying on human resources will consistently result in cost inefficiency and time consumption [4]. Nowadays, barcodes are an ordinarily adopted technique in the retail business for cost reduction, time savings, and minimizing human mistakes [3], [4].

Recently, the barcode system has certain disadvantages; for example, random barcode placement on products and a limited range of barcode readers, which could slow down purchases and impact the shopping experience. RFID (Radio Frequency Identification) was then developed to identify products automatically. Unfortunately, RFID can experience issues when lots of transactions occur because radio waves can interfere with each other [3], [4]. Also, the challenges of significant human involvement in inventory and goods management in supermarkets still remain.

Artificial intelligence (AI) [5] and machine learning (ML) [6] have gained much attention for applying in the different applications. Object detection, as an application of AI, has been developing and can be deployed in the retail industry. These technology advanments have revolutionized various aspects of retail, including on-shelf product identification, automated self-checkout systems, and inventory management. For example, the emergence of automated retail stores like Konzum Smart [7], the first unmanned retail store in Croatia, demonstrates their growing presence in the industry. Konzum Smart uses 150 cameras to track customer movements and record purchases. When customers leave, it automatically calculates the cost and deducts it from their bank account, eliminating the need to queue for payment. These processes increase efficiency, reduce losses, boost profits, achieve inventory, and enhance customer satisfaction.

This paper aims to develop a product recognition model and explore the effectiveness of identifying products on retail store shelves in the modern trade sector using YOLO (You Only Look Once) methodology.

The remainder of this paper is organized as follows: Section 2 describes the related works about object detection for product recognition. Section 3 provides the details on the YOLO algorithm, the datasets, and the evaluation metrics. Section 4 provides the hardware and software configuration, the experiment, and results. Finally, Section 5 presents conclusions, the contribution, and future works.

II. Related work

Many well-known convenience stores, like 7-Eleven, still rely on their employees to verify that products are available on the shelves. Employees must restock the shelves whenever a gap is found. Innovative technology has been used in a variety of areas to solve these issues, with the goals of reducing human error, avoiding sales loss, inventory discrepancies, and maintaining consumer satisfaction.

There are several studies using YOLO [8, 9] techniques in various related industries. For example, in the food sector, sophisticated processes are involved in the production of consumable goods. Fotios K. Konstantinidis et al. [10] explored how to automatically identify dairy products in a production line in real-time to automate quality-related standards and packaging procedures to enhance accuracy and speed in defect detection. YOLOv5 [11] was the chosen framework for their investigation. The experiment was initiated by creating 110 different images. 88 images were used for training and another 22 for validation. The experimental results revealed no significant disparity between the two algorithms, both achieving an impressive accuracy rate of 99%.

Likewise, the use of recycled materials in the food packaging industry has grown during the past several years. However, only a few recycling methods can be used, such as barcodes, which are sometimes worn out, leading to problems. Sree Chandan Kamireddi's research [12] focused on object detection models to create a dataset containing images of Tetra packs with visible recycle logos. Four deep learning models, including Faster-RCNN [13], YOLO, SSD [14], and Mask-RCNN [15], were investigated, which performed best. The dataset was collected from nearby supermarkets and augmented using Roboflow. YOLOv5 came out on top in speed and overall performance compared to the other models. Besides, the food packaging industry, there was an idea to detect a real-time packaging defection system based on deep learning techniques. Thi-Thu-Huyen Vu et al. [16] developed an approach to detect defects in the packaging in real time with the YOLO algorithm using YOLOv5. The data was collected from FFmpeg, the multimedia open-source. Two hundred image files of damaged boxes and another 200 images of intact boxes were captured by a production line. The model accuracy achieved the mAP score of 78.6%.

In the retail sector, many convenience stores currently do not have a well-organized structure for inventory management. This inefficiency creates added expenses for inventory control and storage. The overstock and the shrinkage reflect the inventory availability. This is comparable to studies by Piero Herrera Toranzo et al. [17] that used YOLOv5 to identify, measure, and confirm bottles and canned goods. When the various models were compared and the algorithm's effectiveness was assessed, the findings showed that YOLOv5 produced the most accurate identification, counting, and verification of canned products and bottled goods in supermarkets. In addition to the inventory control issue, recent research, including work by Harish Kumar [18], has focused on developing real-time item detection systems to identify empty shelves. A collection of 28,000 images that included 80 popular grocery items was used in this study. The results confirmed the YOLOv8 algorithm's efficiency in recognizing unoccupied shelf spaces, demonstrating that it could identify more than 50% of the empty shelf area.

Furthermore, another concern is that it is hard to identify inferior categories separately. Products can sometimes only differ slightly in appearance, which makes it difficult for humans to distinguish between them. Jacqueline Abyasa et al. [19] researched the brand recognition of grocery products using YOLOv8, focusing on the inter-class similarities. The dataset, comprising of 3,095 images representing 20 grocery product classes, was internally developed. The dataset was divided into three sets: training (70%), validation (20%), and testing (10%). The results showed that YOLOv8 competed well in recognizing products even when met with inter-class similarities. Subsequently, the outcomes showed that this approach could efficiently automate checkout procedures in smart retail stores.

Nevertheless, YOLOv8, the cutting-edge model, provides speed and reliability. However, previous studies have not focused on comparing its different versions. Additionally, no experiments have been conducted to evaluate how well of YOLOv8 versions for the Grozi-120 dataset, which could help evaluate YOLOv8's performance. The Grozi-120 dataset contains the images of 120 grocery products in both controlled environment (in vitro) and real-world (in situ) settings. This diverse setup provides a valuable tool for evaluating model performance under different conditions.

Finally, previous studies have certain constraints. They struggle in extremely bright or dark lighting conditions, and obstructed views can hinder their ability to detect empty shelves accurately. In addition, factors such as poor image quality, inappropriate camera angles, and low or blurry resolution can reduce the precision of detecting empty shelves. Besides, the standard model evaluation datasets like PASCAL VOC [20], which includes only 20 object classes, and MS COCO [21], with its 80 object categories, are insufficient for directly applying current object detection algorithms to retail product recognition. Fig. 1 illustrates a comparison of the VOC 2012 and COCO dataset results using three algorithms: Faster R-CNN, SSD, and YOLOv2 [22]. The comparison reveals a noticeable decrease in detector accuracy as the number of classes increases.

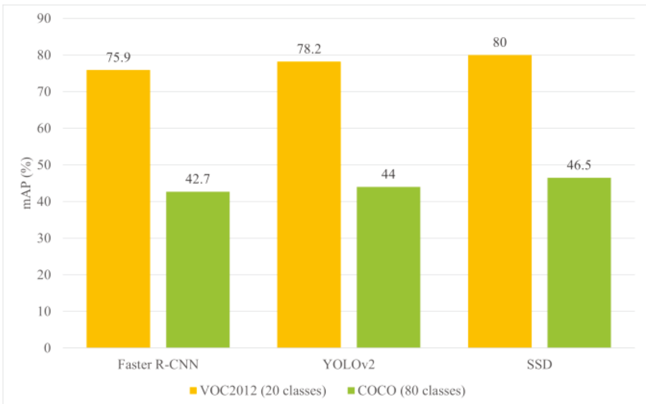

Fig. 1. The comparison result on VOC2012 and COCO datasets with different algorithms [3].

TABLE I. THE RELATED WORKS COMPARISON

No.	Ref./Year	Objectives	Techniques	Dataset	Results
1	[10], 2023	‣ Increase accuracy and identify the speed of detecting the defect ‣ Identify dairy products within a product line in real-time to automate quality related standards and packaging procedures	‣ YOLOv5 ‣ Mark R-CNN	YogDATA (Yogurt cup recognition dataset): 110 images	No noticeable difference between 2 algorithms which is 99% of accuracy.
2	[12], 2022	‣ Create the dataset covering certain pictures of the Tetra packs with visible recycle logos ‣ Test and Compare the performance among the different algorithms	‣ Faster-RCNN ‣ YOLOv5 ‣ SSD ‣ Mask-RCNN	Images of the logo side of the Tetra packs: 130 images from a smartphone	YOLOv5 is the best model in the comparison. (mAP = 0.771)
3	[16], 2023	To introduce a real-time packaging defect detection system	YOLOv5	Packaging defect dataset: 400 images from a production line.	mAP: 78.6%
4	[17], 2023	‣ To detect, count, and verify the status of bottles and canned products in supermarkets ‣ To compare the algorithms efficiency	‣ YOLOv5 ‣ EfficientDet ‣ DET ‣ Faster R-CNN	‣ CanBo-Pe: 1,640 images ‣ CanBo-Pe +Status: 2,400 vitro images	YOLOv5s achieved the mAP50 score of 0.935 and the mAP50-95 score of 0.893.
5	[18], 2023	To develop a model that can identify empty shelves in real-time	‣ YOLOv8 ‣ Roboflow AutoML	‣ GroZi-3.2K: 11,585 images ‣ SKU110K: 11,762 images ‣ WebMarket: 300 images ‣ Grocery Product: 28,000 images, 80 classes	Roboflow AutoML achieved a mAP = 49.5%
6	[19], 2023	To analyze the performance of YOLOv8 for product recognition in grocery store with the inter-class similarity problem	YOLOv8	Internally developed dataset: 3,095 images	YOLOv8 was able to recognize products with inter-class similarity problems, achieving an mAP50-95 score of 0.81.

This research challenge is still the interclass classification with different methods and algorithms. The minor visual distinctions between some grocery products make them difficult to differentiate. Moreover, dissimilar environmental factors can impact the accuracy of product recognition, including background settings, lighting conditions, occlusions, and the way products are arranged.

Table I presents a comparative analysis of prior studies on object detection in various fields.

III. METHODOLOGY

This section describes the method used in this research to explore the ability of detecting products. The methodology is demonstrated in Fig. 2.

A. Dataset Collection

For this paper, the experiment uses the public standard dataset to assist in testing models and comparing the results. This study features two datasets concisely summarized in Table II. Both datasets are retrieved from the Roboflow website.

TABLE II. DATA SPLITTING DESCRIPTIONS

Dataset	Training	Validation	Testing	Total
GroZi-120	7,502	2,323	1,133	10,958
SKU110K	6,998	2,000	1,000	9,998

GroZi-120 [23]: For this dataset, there are 120 grocery classes with variations, such as color, size, and shape. Each item has two types of images: high-quality studio images (in vitro) and natural environment images (in situ). The dataset consists of 10,958 images, divided into 1,133 test images, 7,502 training images, and 2,323 validation images.

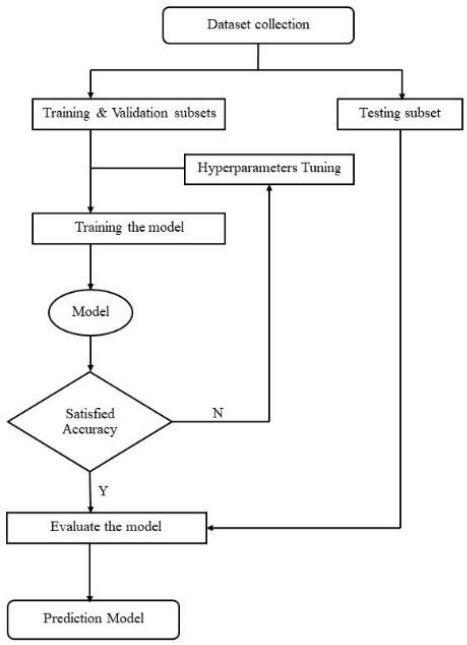

Fig. 2. Research methodology.

SKU110K [24], [25]: This dataset has bounding box annotations for objects found on store shelves, and it consists of a single class named ‘object’. SKU110K contains 9,998 densely packed supermarket shelf images with over 1.7 million annotated bounding boxes. It's split into 6,998 images for training, 2,000 for validation, and 1,000 for testing purposes.

All datasets are imported in YOLO format and sourced from Roboflow. The samples of the image dataset are depicted in Fig. 3.

SKU110K

Grozi-120: in situ Grozi-120: in vitro

Fig. 3. Examples of data images in Grozo-120 and SKU110k dataset.

B. Training, Validation, and Testing subsets

For creating a product recognition model, the data is divided into three distinct datasets: a training set, a validation set, and a test set. The training set represents the set of data that trains the model. It consists of images and labels used to train data. During each epoch, the model will repeatedly be trained on this data in the training set. The validation set is a set of data, separated from the training, used to validate the model during the training. This validation process provides the information used to adjust the hyperparameter settings. These two subset processes will train, validate, and fine-tune hyperparameters.

The test set is a set of data used to evaluate the performance of the model after the model has already been trained. This set is separated from both the training set and the validation set as the unseen data. Figure 3 demonstrates the dataset splitting.

In the scope of this research, the dataset collections are partitioned into a train set, a validation set, and a test set in a 70:20:10 ratio as shown in Fig. 4.

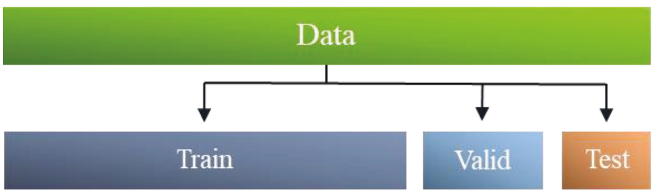

Fig. 4. Dataset splitting.

C. YOLO Models Training

Object detection [26] is an essential application in the domain of computer vision that identifies objects and locates them in images or videos. YOLO, one of the best object detection algorithms, is a single-stage neural network that instantaneously predicts object bounding boxes and class probabilities. It has been developing until now, YOLOv8 [27] which can detect objects faster and more reliability. Thus, in this experiment, YOLOv8 is selected and compared with different YOLOv8 versions to determine the best performance.

The images are divided into smaller pictures for better small object detection using ReLU [28] activation. YOLOv8 utilizes the modified CSPDarknet53 backbone [9], similar to YOLOv5. However, in YOLOv5, the CSPLayer is replaced by the C2f module. Additionally, it incorporates PAN-FPN in the neck process following the extraction by Darknet53.

YOLOv8 also adopts a new anchor-free model with a decoupled head [9] that allows the pattern to reduce the number of box predictions and parameters to speed up the training. Moreover, it uses a new feature pyramid network architecture that improves the feature extraction, better recognize objects of different sizes, and improves accuracy. As a result, it gains more competence and can widely be applied in CPU and GPU.

In this research experiment, the sharing configurations are 150 epochs, a patience value of 75, and a batch size of 16.

D. Evaluation Metrics

For evaluating the effectiveness of detection metrics, Intersection over union (IoU), precision, recall, and mean average precision (mAP) were employed as benchmarks in the evaluation of product recognition. The experiments were conducted based on the confusion matrix. The equations were presented respectively.

$$\text{IoU} = \frac{\text{Area of Overlap}}{\text{Area of Union}} \tag{1}$$

IoU is used to assess the object localization accuracy, measuring the overlap between the ground truth and predicted bounding boxes. The prediction accuracy is determined by the threshold value using IoU.

$$\text{Precision} = \frac{TP}{TP+FP} \tag{2}$$

Precision quantifies how well the ML model correctly predicts the target labels. It describes that when an object exists, the bounding box created by the model is precise in proportion to the precision value. Precision can detect potential accuracy manipulation by evaluating the reliability of positive predictions. The system's goal is to optimize these metrics to make as few mistakes as possible when predicting the positive labels. TN stands for true negatives. TP is for true positives. FN and FP denote a false negative and a false positive, respectively.

$$\text{Recall} = \frac{\text{TP}}{\text{TP+FN}} \quad (3)$$

Recall describes how accurately the ML model identifies true positives from all the actual positive samples within the dataset. It means the ability of the model to accurately identify a portion of the actual ground truths according to recall value and classify them to a particular class. The goal of this system is to find and to try every positive label there is to be found.

$$\text{mAP} = \frac{1}{\text{N}}\sum_{i=1}^{\text{N}} \text{AP}_i \quad (4)$$

The mAP describes the model's accuracy. The high mAP indicates the model gets a low false positive (FP) and a low false negative (FN) rate. The higher the mAP, the better the model precision, and the higher the recall. Improvement of data quality and algorithm optimization can enhance the mAP outcomes.

IV. Experiment and Results

This section describes the experiment of YOLOv8n, YOLOv8s, YOLOv8m, YOLOv8l, and YOLOv8x with the tested datasets. GroZi-120 and SKU110K datasets are deployed. Each model is compared to find the best performance.

This experiment operates on YOLO version 8, Python version 3.10.12, torch version 2.1.0 with CUDA version 12.1 (running on NVIDIA GPU A100-SXM4-40GB, VRAM 40514 MB), Operation System is Linux Ubuntu 22.04.3 LTS (Jammy Jellyfish), running on Google Colab Pro+. Moreover, CPU specifications include Intel (R) Xeon (R) CPU @ 2.00GHz., 8-core CPU, 51 GB of RAM, and 202 GB of data storage capacity.

Table III. Experiment results with Grozi-120 dataset

Grozi-120				
Model	***Precision***	***Recall***	***mAP50***	***mAP50-95***
YOLOv8n	0.270	0.266	0.293	0.288
YOLOv8s	**0.291**	0.268	0.301	0.296
YOLOv8m	0.288	0.286	0.303	0.298
YOLOv8l	0.284	0.285	0.302	0.297
YOLOv8x	0.252	**0.295**	**0.306**	**0.301**

YOLOv8 Performance: the result evaluation is evaluated with four metrics: Precision, Recall, mAP50, and mAP50-95. In this paper, 150 epochs are executed with a patience of 75. Table III demonstrates each YOLOv8 metric performance using the Grozi-120 dataset. YOLOv8s showed the best precision (29.1%) when used with Grozi-120, while YOLOv8x provided the best overall results in Recall (29.5%), mAP50 (30.6%), and mAP50-95 (30.1%).

Table IV illustrates the result of the YOLOv8 performance for each metric using the SKU110K dataset. Still, YOLOv8x shows the best performance in Recall (87.5%), mAP50 (92.6%), and mAP50-95 (59.6%), while YOLOv8l gives the best precision. However, the precision result is not significantly different from other YOLOv8 versions.

Table IV. Experiment results with SKU110K dataset

SKU110K				
Model	***Precision***	***Recall***	***mAP50***	***mAP50-95***
YOLOv8n	0.896	0.833	0.901	0.564
YOLOv8s	0.902	0.854	0.915	0.581
YOLOv8m	0.905	0.866	0.922	0.590
YOLOv8l	**0.907**	0.871	0.925	0.595
YOLOv8x	0.906	**0.875**	**0.926**	**0.596**

The experiment shows that YOLOv8x gets the best scores in overall performance when evaluated with both datasets. The bounding box is precise in proportion to the precision value, with a high detection rate, 90.6%. Additionally, over 80% of relevant elements can be detected. The model's accuracy varies with the IoU threshold: 92.6% at 0.5 and 59.6% at thresholds between 0.5 and 0.95. Tables 3 and 4 demonstrate a correlation between model size and accuracy; larger models perform better. However, this increased accuracy comes with the trade-off of slower execution and larger storage needs.

V. Conclusion

This research paper applied the different versions of YOLOv8 algorithms (nano, small, medium, large, and extra-large) for Grozi-120 and SKU110K datasets. The evaluation metrics are precision, recall, mAP50, and mAP50-95. The outcome shows that YOLOv8 can detect retail products on shelves with a high model accuracy of 92.6% for SKU110K and an accuracy of 30.6% for the Grozi-120 dataset. Based on the results, YOLOv8x achieves the best performance. However, Grozi-120 performed poorly due to its large dataset and there are many classes. This led to lower mAP50 and mAP50-95 scores. In contrast, the single-class SKU110K dataset yielded higher accuracy and mAP scores. This demonstrates an inverse relationship between the number of dataset classes and model accuracy: the accuracy decreases as the dataset complexity increases (more classes). Moreover, the evaluation indicates a positive correlation between model size and accuracy, with larger models consistently performing better. Finally, according to the experimental results, YOLOv8x has the ability to be applied in a real-world object detection algorithm for retail product recognition and to be beneficial to the retail industry.

For the contribution of this paper, the experiment has explored the performance of YOLOv8 versions for Grozi-120 and SKU110K datasets since the previous research papers have never been done before. In future research, more datasets and different YOLO models should be experimented and evaluated to provide more analysis results.

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